Fiscal Year:	FY 2019	**Task Last Updated:**	FY 01/06/2019
PI Name:	Long, Marshall Ph.D.		
Project Title:	Fuel Dilution Studies of Coflow Laminar Diffusion Flames in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	marshall.long@yale.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-432-4229
Organization Name:	Yale University		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	PO Box 208284		
PI Web Page:			
City:	New Haven	**State:**	CT
Zip Code:	06520-8284	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	03/01/2017	**End Date:**	02/28/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Period of performance changed to 3/1/2017-2/28/2020 (original POP was 1/1/2017-12/31/2019) per D. Stocker/GRC (Ed., 1/17/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smooke, Mitchell Ph.D. (Yale University)		
Grant/Contract No.:	NNX17AD49A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Coflow Laminar Diffusion Flame" grant NNX11AP43A, with investigators Drs. Marshall Long and Mitchell Smooke. Diffusion flames are the flame type of most practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in such systems is critical in the modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place. Turbulent diffusion flames are generally nonbuoyant due to the large flow velocities. In addition, in practical devices combustor configuration and flow field patterns may produce flames with sharp edges. Extinction along these edges can result from incomplete combustion. In other regions combustion generated soot particulates may be formed. To understand better the factors that affect flame extinction and particulate formation, we propose to focus this study on flame phenomena at the extremes of the fuel dilution spectrum – namely dilute flames near extinction and pure-fuel (sooting) flames. Previous work has shown that existing computational models are unable to predict accurately the behavior of these systems. The enhanced time and spatial scales in a microgravity environment will enable the stabilization of flames with increased levels of dilution and will help generate flames that have increased soot residence times compared to normal gravity facilities. This should provide an excellent platform to gain a better understanding of the factors that affect diffusion flame lift-off distance and extinction, and it should also provide an excellent test-bed to determine the roles of inception, surface growth, and oxidation processes in soot formation models. The overall goals of the research are to improve our understanding of the physical and chemical processes controlling diffusion (i.e., non-premixed) flame structure and lifting phenomena (i.e., stabilization) and to provide for rigorous testing of numerical models, including thermal radiation, soot formation, and detailed chemical kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Combustion of fossil fuels is responsible for the majority of energy production in the world today and provides the power for transportation systems of all kinds. Effective modeling of combustion processes is critical to improving efficiencies and reducing pollutants in practical combustion devices. The ability to predict the coupled effects of complex transport phenomena with detailed chemical kinetics in diffusion flames will improve modeling of turbulent reacting flows and in understanding the processes by which extinction, soot formation, and radiative transfer take place.
Task Progress:	Microgravity Experiments: CLD Flame Round One The Coflow Laminar Diffusion (CLD) Flame experiment, which initially began November 2017, continued operations with round one of testing going into the new year and was completed at the end of February. During round one, CLD Flame saw 19 test days, 12 of which occurred in 2018. The tests consisted of methane/air flames at extreme dilution limits as well as pure fuel sooting conditions. The initial tests from the months November and December 2017 found challenges with the Advanced Combustion via Microgravity Experiments (ACME) flow and ignition hardware that needed to be addressed before the desired data could be obtained. Starting in January, CLD Flame and the NASA Glenn ops team employed new strategies to meet these challenges. An optimized igniter position was found to improve the success rate of flame ignition attempts. Improved mass flow controller (MFC) calibration settings and increased flow times during testing were used to improve the stability and predictability of the gas flows during flame measurements. A subsequent challenge related to the gas flow issues was the excessive use of air during experimentation. CLD Flame included several test points to determine if nitrogen, rather than air, could be used as the ambient chamber gas; it was found that the differences were indiscernible. Thus, nitrogen was used for the majority of CLD Flame testing as the chamber gas to better manage the amount of daily air use. Highly diluted methane flames were tested as low as 25/75% methane/nitrogen. However, at this extreme condition, the flames were very weak and found to be very susceptible to blow out from flow instabilities. Because issues persisted with the MFC performance during flow transition stages, the CLD Flame team was unable to determine the dilution limit of these microgravity flame. Discussions between the investigating team and the NASA Glenn ops team resulted in the conclusion that a third round of CLD testing will be planned following the completion of other scheduled ACME tests. The task of determining the dilution limit would be a primary task of CLD Flame round three. Flame images were taken of steady, diluted methane flames from 30/70, 40/60, and 50/50% methane/nitrogen. At these concentration conditions, no visible appearance of soot was found. At 60/40%, a slight bit of soot was starting to become visible and quickly transitioned to very sooty flames at the 80/20 and 100/0% conditions. Using the thin-filament pyrometry (TFP) fiber array, flame temperature measurements were taken at 30/70 and 40/60% at various flow velocity conditions. For the 80/20 and 100/0% concentration conditions, soot temperature measurements were taken from the unsaturated flame images. From the flame images, CH* number density and soot volume fraction can be calculated but require an absolute light calibration. The absolute light calibration was done using the radiative properties of the heated TFP fibers. Multiple test points could be used to perform this light calibration. The verification of the calibration is an on-going process. Microgravity Experiments: CLD Flame Round Two CLD Flame testing resumed for round two at the end of May of 2018 and continued up to September. During this period, CLD Flame saw 17 days of operation. This round of tests consisted of obtaining measurements of highly nitrogen-diluted and heavily sooting ethylene/air flames. This followed round one of the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames). At the beginning of round two, it was immediately apparent that a significant amount of soot had accumulated on the coflow burner. The accumulation occurred due to an error in the input parameters of one of the last E-FIELD Flames test points that caused an augmented flame soot production and subsequently the electric field caused the soot to migrate towards the burner. This accumulation of soot on the burner was originally overlooked during the transition between E-FIELD Flames and CLD Flame. Due to concerns that the soot would compromise the flow boundary conditions, the coflow burner was taken from the Combustion Integrated Rack (CIR) and cleaned, removing all soot from the inside of the fuel tube and the coflow honeycomb. Following the coflow burner cleaning, CLD Flame began with sooting flame images. As expected, the ethylene flames were extremely sooty and much taller than the comparable methane flames. In round one, the ACME digital color camera settings were all constant except for the exposure time. With these taller and brighter ethylene flames, new camera settings were required, and thus, new spatial and light calibrations were necessary. Multiple calibration test points were performed throughout both rounds one and two. Repeating the calibration over multiple days allowed for

the characterization of the ACME camera's performance over time.

Flame images were captured of ethylene/air flames with concentrations of 100/0, 80/20, 60/40, 50/50, 40/60, 30/70, 20/80, 15/85% ethylene/nitrogen. Soot remained visible in these flames until the concentration reached as low as 20/80%. Similar to the methane/air flames in round one, the soot temperature and soot volume fraction could be calculated from the sooting flames. But unlike the non-sooting methane flames, CH* could not be calculated from the ethylene flames at 20/80 and 15/85 since these flames had a greener hue which indicated that more than CH* chemiluminescence was being emitted from the flames. The TFP fiber array was used to measure flame temperature profiles in the non-sooting ethylene flames at 20/80 and 15/85%. The measurements were made at various gas velocity conditions where the coflow and fuel mixture velocities were both matched and mismatched to study the effect of the coflow air stream.

Microgravity Experiments: Planning for Round Three

During the first two rounds of CLD Flame testing, poor performance of the MFCs during the transition stages between changing velocities persisted. This prevented the investigation of extreme dilution limits of microgravity flames for both the methane and ethylene flames. It was unclear whether the extremely weak flames were blowing out because it reached its extinction limit or blown out due to the unpredictable behavior of the nitrogen and air MFCs. Discussions with the NASA Glenn ops team led to the decision that extreme dilution limits should be the primary goal of a third round of CLD Flame tests. Because the soot volume fraction and CH* number density calculations rely heavily on the absolute light calibration, additional light calibration testing will also be required to ensure the performance of ACME's digital color camera is characterized and maintained over the time period between the conclusion of round two and the beginning of round three.

Post Microgravity Experiments: Data Analysis

Since the beginning of ACME International Space Station (ISS) testing, the CLD Flame team has made advancements in processing the ISS data as it was being generated. Many of these results have been presented at conferences throughout the year.

The flame images of methane and ethylene flames have been processed to characterize flame geometry and lift-off height. For methane flames, preliminary calculations of CH* mole fractions have been made of Abel-inverted flame images. Temperature profiles were generated from TFP fiber sweep tests on highly nitrogen-diluted flames at velocities ranging from 20 to 40 cm/s. The same was done for nitrogen-diluted ethylene flames, with velocities ranging from 30 to 50 cm/s. Most work has been focused on these fiber temperature measurements since the light calibration is dependent on them. Preliminary measurements have also been made of soot temperature and soot volume fractions from heavily sooting methane and ethylene flames.

Ground-Based Experiments

In the final months of 2018, work has begun on configuration of a Yale-based ground copy of the ACME CLD Flame experiment. The preparation includes reconfiguring the hardware to match more closely to the final ISS test point conditions and procedures. The ground copy testing will include flame images and TFP measurements of both methane and ethylene flames. Absolute light calibration tests will also be performed with the ground copy to allow quantitative measurements of CH* concentrations and soot volume fractions. In addition to normal gravity tests on the ground copy of the ACME CLD Flame burner, the color data camera will be used for soot temperature and volume fraction measurements on the Yale standard coflow burner. This will provide even further verification of the analysis methods by comparing to measurements done on a burner that has already been well characterized for many years through multiple diagnostic techniques.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Tinajero J, Giassi D, Stocker D, Smooke M, Long M. "Experimental Study of Highly Diluted Methane Coflow Flames in Microgravity." Work-in-progress poster for the 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. http://www.combustionsymposia.org/2018/loadpage/program/wipp-abstracts , Aug-2018
Abstracts for Journals and Proceedings	Tinajero J, Giassi D, Stocker D, Smooke M, Long M. "Experimental Study of Sooting Methane Coflow Flames in Microgravity." Work-in-progress poster for the 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. 37th International Symposium on Combustion, Dublin, Ireland, July 29-August 3, 2018. http://www.combustionsymposia.org/2018/loadpage/program/wipp-abstracts , Aug-2018
Abstracts for Journals and Proceedings	Tinajero J, Giassi D, Stocker D, Long M. "Experimental Study Highly Diluted and Sooting Methane Coflow Flames in Microgravity." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Nov-2018
Articles in Peer-reviewed Journals	Kempema NJ, Long MB. "Effect of soot self-absorption on color-ratio pyrometry in laminar coflow diffusion flames." Opt Lett. 2018 Mar 1;43(5):1103-6. https://doi.org/10.1364/OL.43.001103 ; PubMed PMID: 29489790 , Mar-2018